

Campus Recruitment Prediction

1 Introduction

In this assignment I developed a classification model to predict whether a student will be placed during campus recruitment, using academic, demographic, and institutional features. What I did was to test multiple classification models, compare their performance, and identify the most effective approach.

2 Data

2.1 Dataset

The Campus Recruitment Dataset contains records from placement drives with features including academic scores, demographic details (e.g., gender, work experience), institutional attributes, and the target variable: placement status. It has 215 rows and 15 columns representing the features.

	s1_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t	workex	etest_p	specialisation	mba_p	status	salary	
	0	1	M	67.00	Others	91.00	Others	Commerce	58.00	Sci&Tech	No	55.0	Mkt&HR	58.80	Placed	270000.0
	1	2	M	79.33	Central	78.33	Others	Science	77.48	Sci&Tech	Yes	86.5	Mkt&Fin	66.28	Placed	200000.0
	2	3	M	65.00	Central	68.00	Central	Arts	64.00	Comm&Mgmt	No	75.0	Mkt&Fin	57.80	Placed	250000.0
	3	4	M	56.00	Central	52.00	Central	Science	52.00	Sci&Tech	No	66.0	Mkt&HR	59.43	Not Placed	NaN
	4	5	M	85.80	Central	73.60	Central	Commerce	73.30	Comm&Mgmt	No	96.8	Mkt&Fin	55.50	Placed	425000.0
	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
	210	211	M	80.60	Others	82.00	Others	Commerce	77.60	Comm&Mgmt	No	91.0	Mkt&Fin	74.49	Placed	400000.0
	211	212	M	58.00	Others	60.00	Others	Science	72.00	Sci&Tech	No	74.0	Mkt&Fin	53.62	Placed	275000.0
	212	213	M	67.00	Others	67.00	Others	Commerce	73.00	Comm&Mgmt	Yes	59.0	Mkt&Fin	69.72	Placed	295000.0
	213	214	F	74.00	Others	66.00	Others	Commerce	58.00	Comm&Mgmt	No	70.0	Mkt&HR	60.23	Placed	204000.0
	214	215	M	62.00	Central	58.00	Others	Science	53.00	Comm&Mgmt	No	89.0	Mkt&HR	60.22	Not Placed	NaN

2.2 Features

There are 15 feature columns.

The features are as follows:

0	sl_no	215 non-null	int64
1	gender	215 non-null	object
2	ssc_p	215 non-null	float64
3	ssc_b	215 non-null	object
4	hsc_p	215 non-null	float64
5	hsc_b	215 non-null	object
6	hsc_s	215 non-null	object
7	degree_p	215 non-null	float64
8	degree_t	215 non-null	object
9	workex	215 non-null	object
10	etest_p	215 non-null	float64
11	specialisation	215 non-null	object
12	mba_p	215 non-null	float64
13	status	215 non-null	object
14	salary	148 non-null	float64

Here we see the number of missing values for each col. The only column with missing values is the salary column with 67 missing values. I handled them by replacing them with mean.

Missing values before being handled:

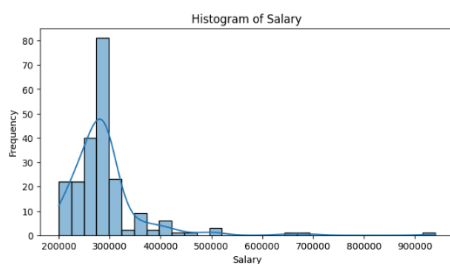
sl_no	0
gender	0
ssc_p	0

ssc_b	0
hsc_p	0
hsc_b	0
hsc_s	0
degree_p	0
degree_t	0
workex	0
etest_p	0
specialisation	0
mba_p	0
status	0
salary	67

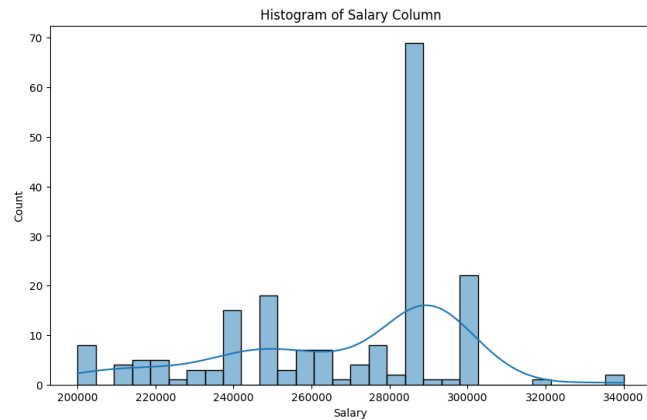
There are 25 outliers which I removed.

2.3 visualization

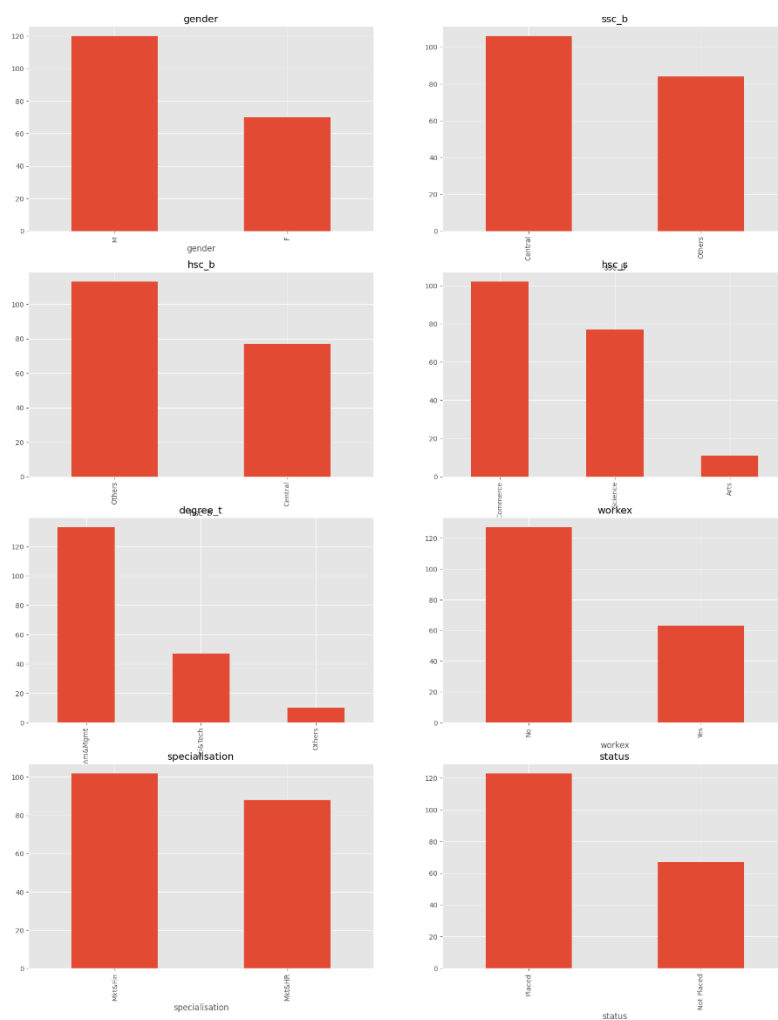
In this section we will go through some visualization related to the data that can help us to have a better sense of our data and the relationship between the features.



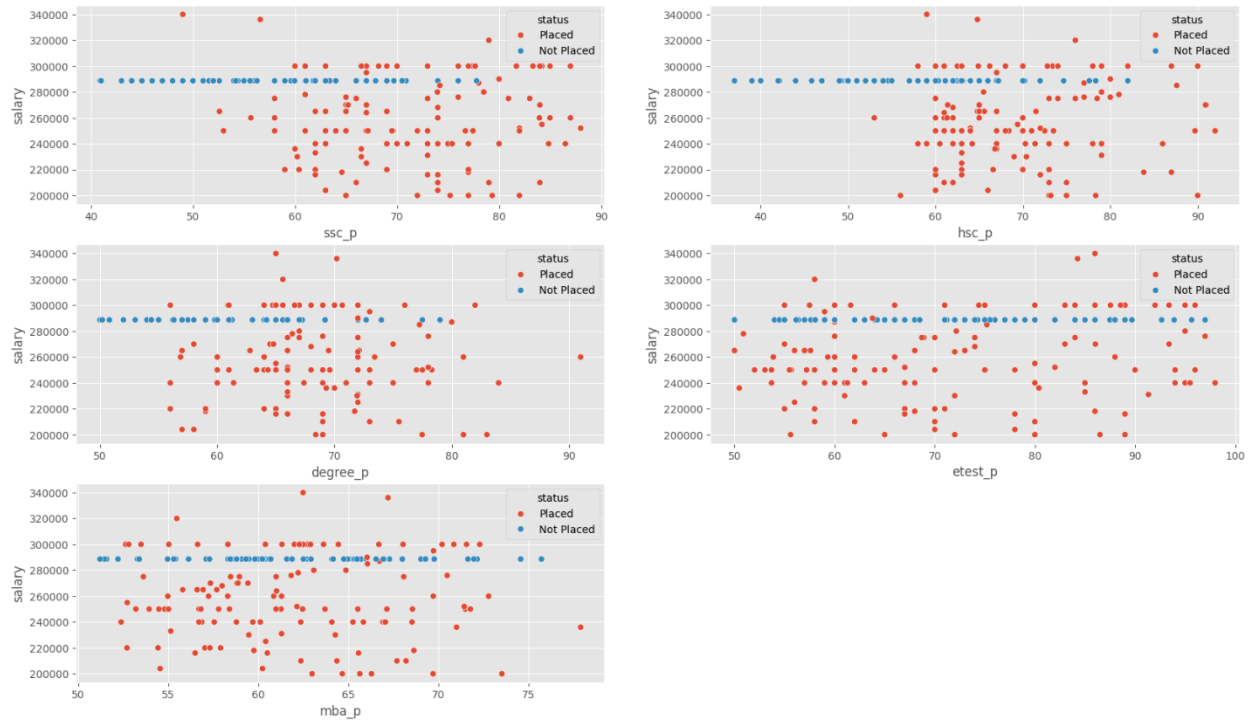
The histogram reveals a right-skewed salary distribution where the majority of people earn between 200,000 and 300,000. As salaries rise the number of individuals earning those amounts drops quickly. Very high salaries showing a typical pattern of income inequality with most earnings clustered in the lower to mid-range. This plot shows the distribution before handling the outliers.



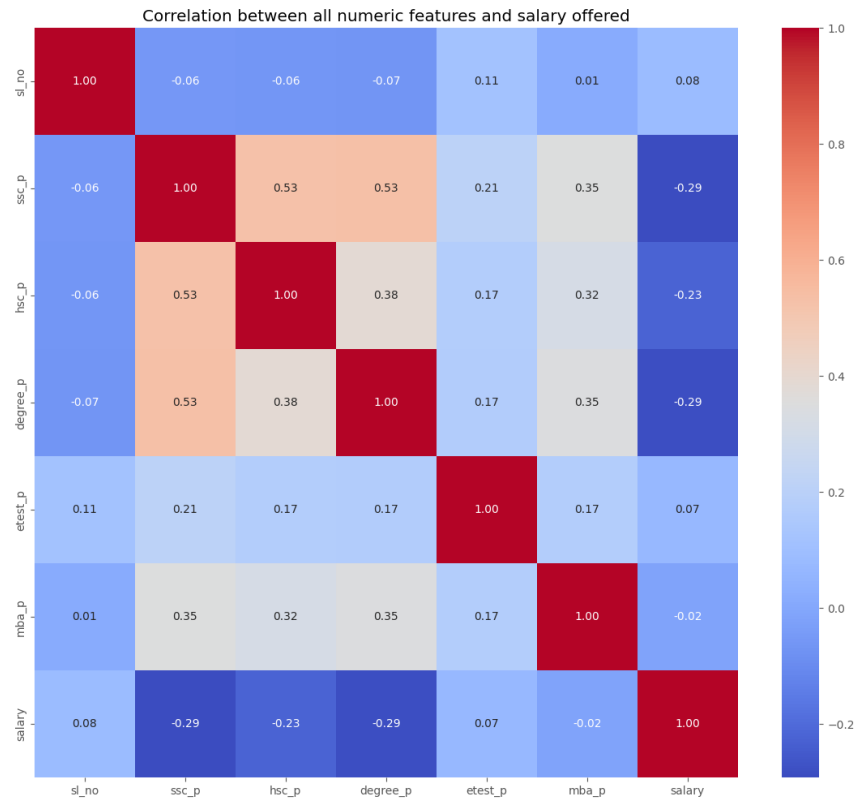
Here we have the distribution of salary after removing the outliers. This pattern suggests that most people in the dataset earn on the lower end of the range with very few high earners. The distribution is clearly right-skewed reflecting a concentration of lower salaries and a long tail of higher less common incomes.



This image shows a set of bar charts or histograms representing categorical variables, likely illustrating the distribution of attributes such as gender, education level, or job roles.



The scatter plots show how salary relates to different academic scores like high school degree and MBA percentages while also distinguishing between students who were placed and those who weren't. Placed students (shown in blue) mostly earn around 300000 with little variation. In contrast not placed students (in red) have more scattered salary values likely due to missing or zero salary entries. Overall there doesn't seem to be a clear link between higher academic performance and higher salary among those who were placed suggesting salaries are fairly standard once placement happens.



The heatmap highlights how salary relates to different academic metrics. There's a slight negative correlation between salary and scores like ssc_p hsc_p and degree_p meaning students with higher marks didn't necessarily get better salaries. Scores from etest_p and mba_p show almost no connection to salary at all. On the other hand academic scores tend to be positively related to each other students who did well in one area often performed well in others especially between secondary higher secondary and degree percentages.

3 Model Development and Evaluation

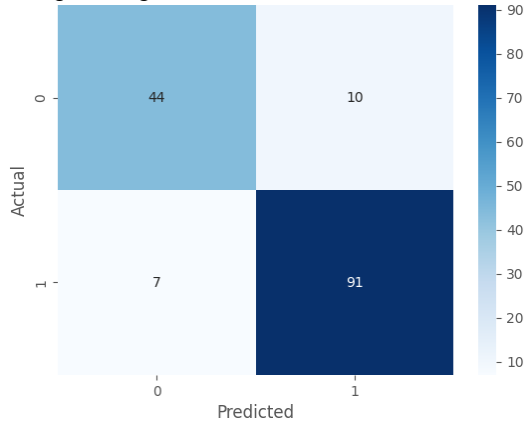
In this section I applied classification models on the data, which is already splitted into 80% train and 20% test, to predict placement.

The following models are applied and their performances are compared : Logistic Regression, Naive Bayes, Linear Discriminant Analysis (LDA)

The evaluation metrics are as follows: Accuracy, Precision, Recall, F1-score, ROC-AUC

3.1 Logistic Regression:

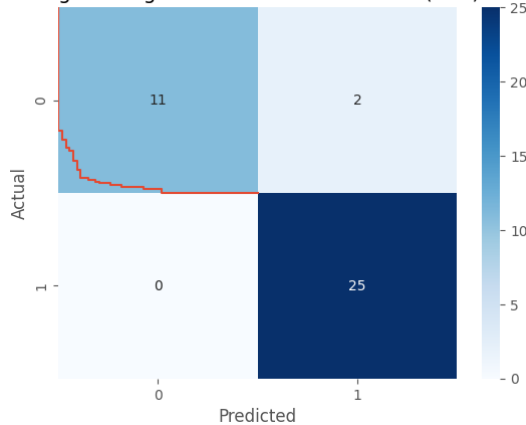
Logistic Regression - Confusion Matrix (Train)



Train Metrics:

Accuracy: 0.8882
Precision: 0.9010
Recall: 0.9286
F1-Score: 0.9146
ROC-AUC: 0.9571

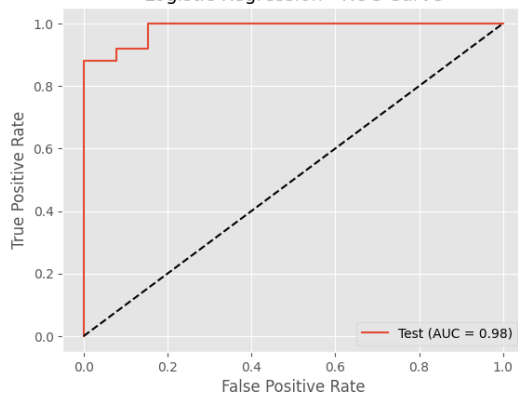
Logistic Regression - Confusion Matrix (Test)



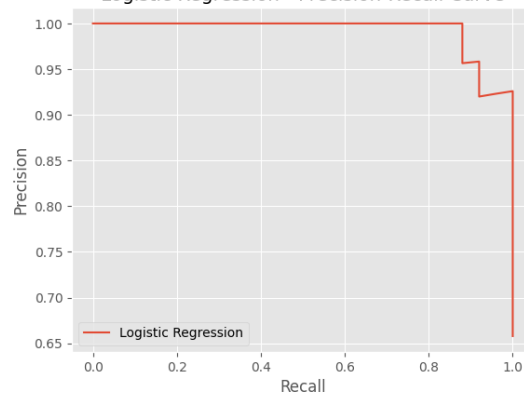
Test Metrics:

Accuracy: 0.9474
Precision: 0.9259
Recall: 1.0000
F1-Score: 0.9615
ROC-AUC: 0.9846

Logistic Regression - ROC Curve



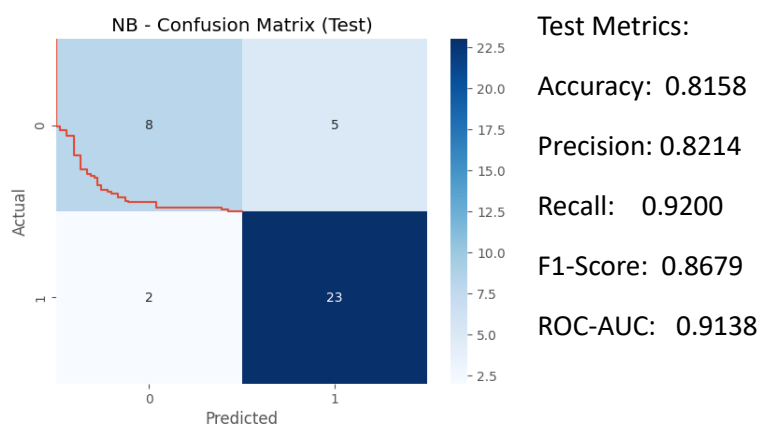
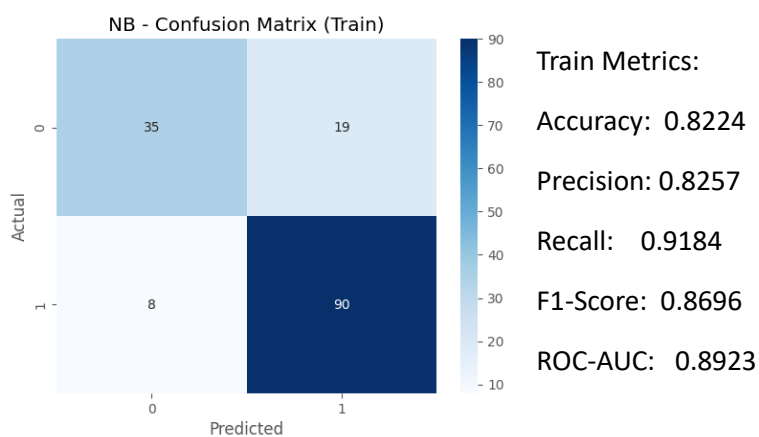
Logistic Regression - Precision-Recall Curve

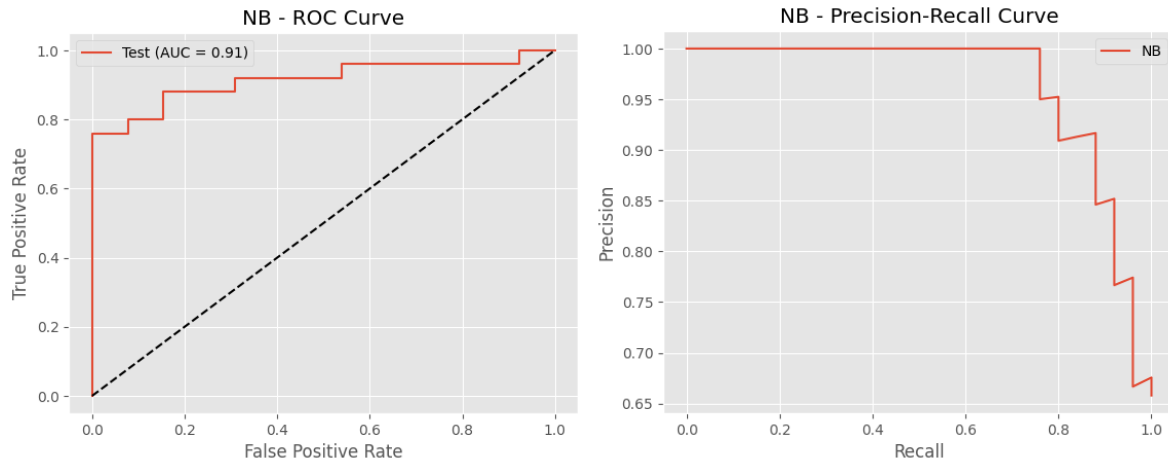


The ROC curve for the logistic regression model shows excellent performance with an AUC of **0.98** indicating near-perfect discrimination between classes as the curve likely hugs the top-left corner. The high AUC suggests the model effectively minimizes false positives while maximizing true positives. Meanwhile the precision-recall curve evaluates class imbalance handling where precision (positive predictive value) and recall (sensitivity) are plotted.

these results are highly suspicious; an AUC of 0.98 and precision consistently near 1.0 for a logistic regression model on a test set are unusually perfect for real-world scenarios. Such exceptionally high performance strongly suggests potential issues.

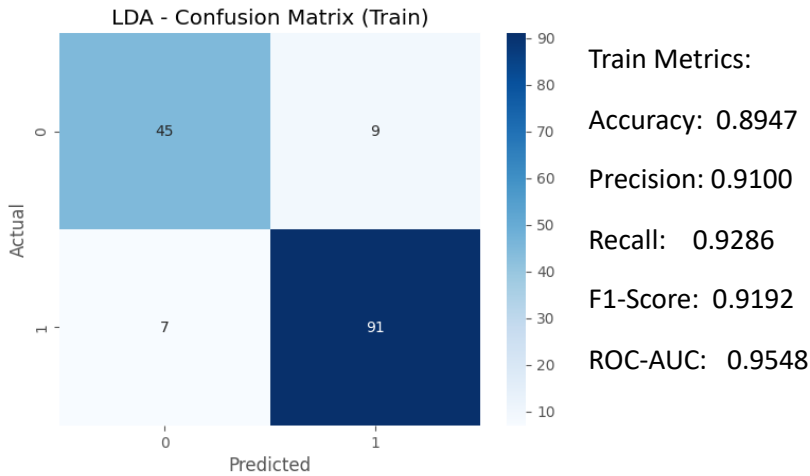
3.2 NB:

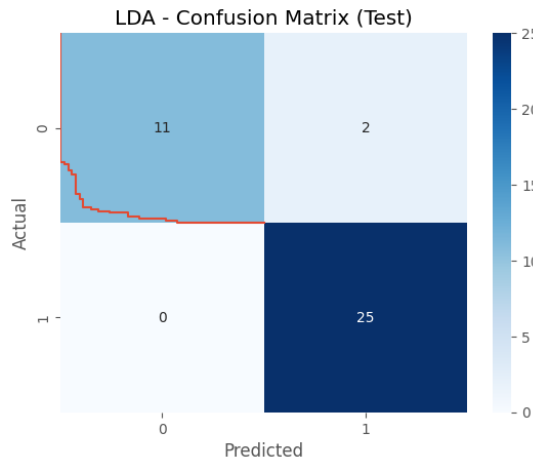




The ROC curve for the Naive Bayes model demonstrates strong classification performance with an AUC of 0.91 indicating good discrimination between classes. While slightly lower than the Logistic Regression's 0.98 this still represents robust predictive ability. The Precision-Recall curve shows near-perfect precision at lower recall levels suggesting highly accurate positive predictions when the model is confident. As recall increases precision gradually decreases reflecting the expected trade-off. These results confirm the Naive Bayes classifier performs well for this binary classification task offering reliable predictions though not quite matching the exceptional performance of the Logistic Regression model. Both curves indicate the model is effective with particularly strong precision at appropriate thresholds.

3.3 LDA:





Test Metrics:

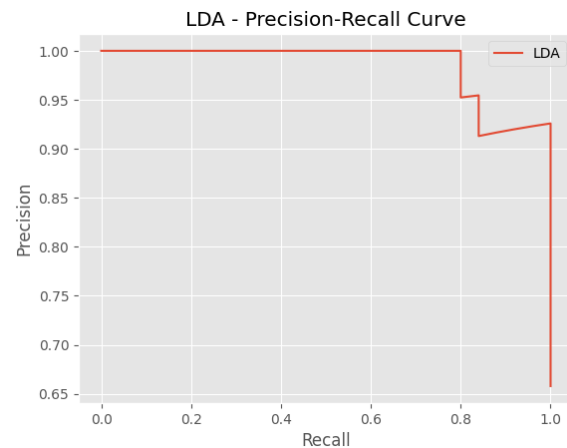
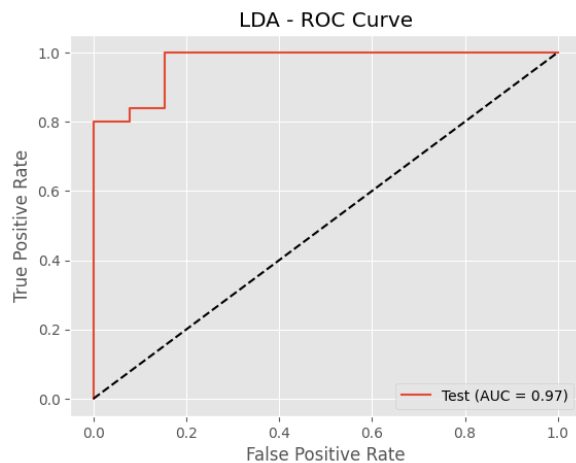
Accuracy: 0.9474

Precision: 0.9259

Recall: 1.0000

F1-Score: 0.9615

ROC-AUC: 0.9723



The LDA model delivers excellent classification performance with an AUC of 0.97 showing near-perfect class discrimination. Its precision remains consistently high across most recall levels confirming reliable positive predictions. These results demonstrate LDA's effectiveness as a robust classifier for this task.

An AUC of 0.97 and a Precision-Recall curve that stays so high for a significant portion of recall are exceptionally good results for an LDA model, or indeed most classifiers in real-world scenarios. This level of performance strongly suggests potential issues.

the problem here is that in Log Reg and LDA for test the Recall is 1. A recall of 1 means that the model correctly identified all actual positive cases—in this context, it predicted every student who was truly "Placed" as "Placed." While this sounds ideal, it's often a red flag because it can indicate that the model is overpredicting the positive class, possibly labeling nearly all cases as "Placed" just to avoid missing any, which leads to many false positives and low precision. This behavior suggests the model is not truly learning meaningful patterns, especially in imbalanced datasets, and may not generalize well to unseen data. One another issue is that our data is not large enough so the model performance can not be expected to be as well as a model with more data.

Best Performing Model: Linear Regression and LDA both have the same accuracy, precision, recall, and F1-score on the test set, with Linear Regression achieving a slightly higher ROC-AUC (98.46% vs. 97.23% for LDA). While Linear Regression performs slightly better, LDA is often preferred over Linear Regression for classification tasks because it works better with categorical labels and is more robust in many cases, especially when the assumptions of the model hold.

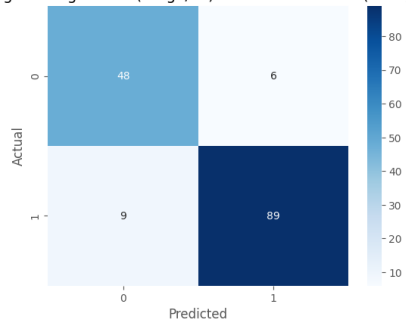
4 Performance Enhancement

In here I applied logistic regression with L1 (Lasso) and L2 (Ridge) regularization using cross-validation to automatically select the best regularization strength and includes `class_weight='balanced'` to correct for class imbalance, which is often the cause of unrealistically high recall (like 1.0).

The `class_weight='balanced'` parameter tells the model to penalize misclassification of minority and majority classes proportionally to their frequency, helping it avoid simply predicting the dominant class (e.g., always "Placed"). By doing so, the model becomes more sensitive to both classes, leading to more realistic recall and precision values, and better generalization. The evaluation function reports key metrics (accuracy, precision, recall, F1, ROC-AUC) for both training and testing sets.

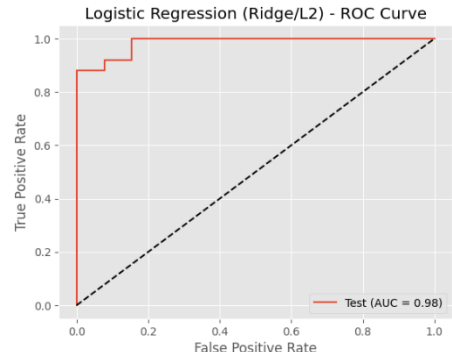
Logistic Regression (Ridge/L2)

Logistic Regression (Ridge/L2) - Confusion Matrix (Train)

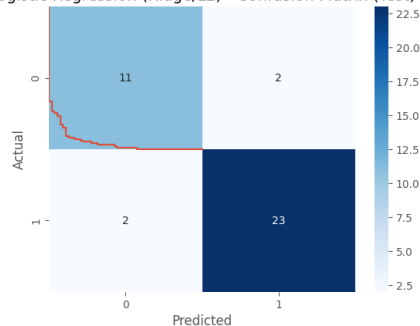


Train Metrics:

Accuracy: 0.9013
Precision: 0.9368
Recall: 0.9082
F1-Score: 0.9223
ROC-AUC: 0.9558

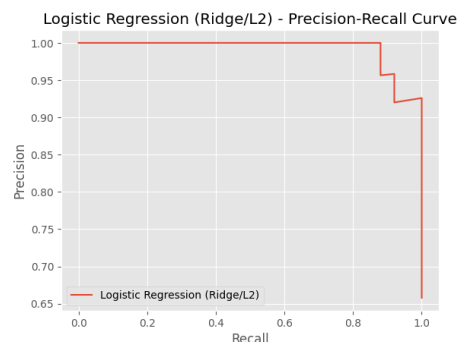


Logistic Regression (Ridge/L2) - Confusion Matrix (Test)



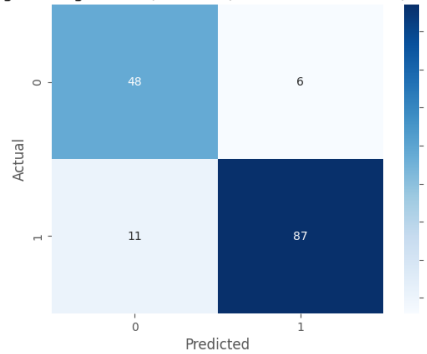
Test Metrics:

Accuracy: 0.8947
Precision: 0.9200
Recall: 0.9200
F1-Score: 0.9200
ROC-AUC: 0.9846



Logistic Regression (Lasso/L1)

Logistic Regression (Lasso/L1) - Confusion Matrix (Train)



Train Metrics:

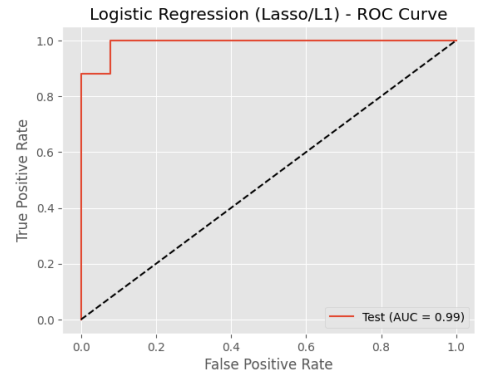
Accuracy: 0.8882

Precision: 0.9355

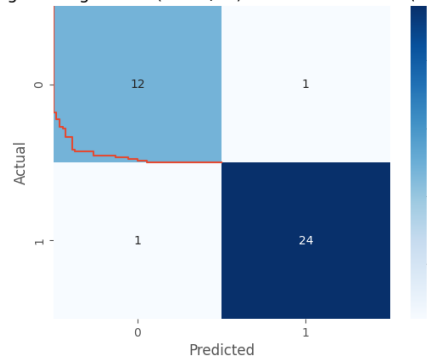
Recall: 0.8878

F1-Score: 0.9110

ROC-AUC: 0.9562



Logistic Regression (Lasso/L1) - Confusion Matrix (Test)



Test Metrics:

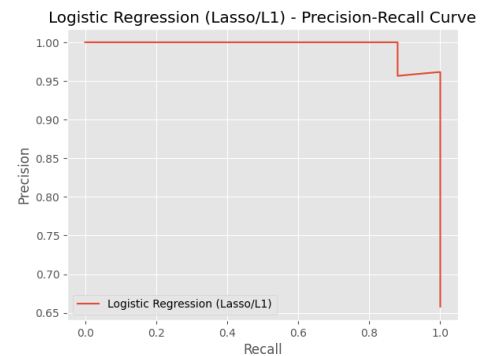
Accuracy: 0.9474

Precision: 0.9600

Recall: 0.9600

F1-Score: 0.9600

ROC-AUC: 0.9908



5 Result

In this comparison of classification models Logistic Regression with Lasso (L1 regularization) emerges as the best performer achieving high metrics across the board with an accuracy of 94.74% precision and recall both at 96% and a perfect F1-score of 0.96. This model also boasts the highest ROC-AUC score of 0.9908 indicating strong overall performance. Logistic Regression without regularization follows closely with similar results also showing excellent precision and recall. LDA also performs very well mirroring Logistic Regression's performance but with a slightly lower ROC-AUC. Naive Bayes while still useful lags behind with lower accuracy (81.58%) and F1-score (0.86) highlighting its weaker performance likely due to its assumption of feature independence. Logistic Regression with Ridge (L2 regularization) performs similarly to the standard logistic model but with a slight drop in accuracy and F1-score suggesting that L2 regularization may be too aggressive for this dataset. Therefore Logistic Regression with Lasso (L1) stands out as the top-performing model offering a balanced and reliable solution.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.9474	0.9259	1.000	0.9615	0.9846
Naive Bayes	0.8158	0.8214	0.920	0.8679	0.9138
LDA	0.9474	0.9259	1.000	0.9615	0.9723
Logistic (Lasso/L1)	0.9474	0.9600	0.960	0.9600	0.9908
Logistic (Ridge/L2)	0.8947	0.9200	0.920	0.9200	0.9846

Conclusion

In this project, I set out to predict whether a student would be placed during campus recruitment using a mix of academic, demographic, and institutional data. After cleaning the dataset—by handling missing values, removing outliers, and exploring the relationships between features—I was able to build a solid foundation for modeling.

I tried out three classification models: **Logistic Regression**, **Naive Bayes**, and **Linear Discriminant Analysis (LDA)**. At first, both Logistic Regression and LDA showed surprisingly perfect results, like a recall of 1 and very high AUC scores. While this might sound great, it raised red flags—these kinds of results usually point to overfitting or issues with class imbalance, especially when working with small datasets.

To fix this, I turned to **regularized logistic regression** with both L1 (Lasso) and L2 (Ridge) penalties, and used **class weight balancing** to help the model treat both placement classes fairly. This made a big difference. The best-performing model was **Logistic Regression with L1 regularization**, which achieved **94.74% accuracy** on the test set, with both **precision and recall at 96%**, and a **ROC-AUC of 0.9908**—a great balance of accuracy and generalization.

Overall, this project showed how powerful regularization and proper handling of imbalanced data can be, especially when dealing with smaller datasets. In the future, bringing in more data and trying out ensemble models like **Random Forests** or **Gradient Boosting** could help push performance even further.

References

[1] <https://www.kaggle.com/datasets/benroshan/factors-affecting-campus-placement/data>

Maryam Soleimani 401222075